Shardul Chogale

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PROFESSIONAL SUMMARY

Computer Engineering student at VIT Mumbai (CGPA 9.32) with hands-on experience building backend APIs, OCR/ML pipelines, and cloud-integrated systems. Shipped real projects using FastAPI, Supabase, and Google Cloud — from a government form digitisation pipeline to an AI plant identification backend. Pursuing dual Azure certifications (AZ-900 & DP-900). Active hackathon participant and community tech leader.

EDUCATION

B.Tech — Computer Engineering — *Vidyalankar Institute of Technology (VIT), Mumbai* Sept 2024 – May 2028
CGPA: 9.32 / 10 | Relevant Coursework: DSA, DBMS, Cloud Computing, Software Engineering, Operating Systems

TECHNICAL SKILLS

Languages: Python, SQL, Java, C
Frameworks & Tools: FastAPI, Flask, REST APIs, Git, GitHub, Data Modeling, System Design
Data Engineering: Data Pipelines, ETL/ELT, SQL Optimization, Real-Time Data Processing
Cloud & Big Data: Microsoft Azure (AZ-900 & DP-900 — In Progress), Apache Kafka, Apache Spark
Databases: Supabase, SQL Databases, Database Optimization, Query Performance Tuning
AI / ML: Google Cloud Vision API, CLIP/ResNet-50, OpenCV, Tesseract OCR, PlantNet API, Gemini API, n8n, scikit-learn, TensorFlow/Keras, PyTorch
Core CS: DSA, OOP, DBMS, Operating Systems, Scalable Systems, Data Flow Architecture

PROJECTS

PaperTrail — Government Form Digitisation Pipeline | *FastAPI, OpenCV, Google Cloud Vision, Firebase, CLIP/ResNet-50*
March 2026

- Built the FastAPI backend handling form ingestion, OCR orchestration, uncertainty scoring, and human verification endpoints; co-designed system architecture covering 2 user roles and a 5-layer pipeline
- Engineered end-to-end OCR/ML pipeline: OpenCV preprocessing (deskew, CLAHE, adaptive threshold) → CLIP template matching → Google Cloud Vision extraction → per-field confidence scoring
- Implemented uncertainty scoring to surface low-confidence fields for operator review before final submission, reducing silent extraction errors
- Designed Firebase storage and audit layer retaining original image, pre-edit extraction, and final corrected form for full traceability — 24-hour SDG 16 hackathon project

GreenGuard — Plant Adoption Platform Backend | *FastAPI, Supabase, PlantNet API, REST APIs* 2025

- Designed and built the complete REST API backend for a plant adoption platform — covering auth, plant management, adoption workflows, community feed, growth reports, and AI identification (25+ endpoints)
- Developed AI plant identification flow: image → PlantNet API → confidence gate (>=30%) → Supabase cache lookup → enriched structured response; caching by scientific name eliminated redundant API calls
- Built robust error fallback system: API failures prompt text-input fallback, low-confidence responses surface top 3 candidates, malformed outputs retry once before returning a graceful generic message

AI-Based Memory Usage Forecaster | *Python, scikit-learn, TensorFlow/Keras, PyTorch, FastAPI, psutil* 2025

- Built an ML system that forecasts per-application memory demand in real time and automatically adjusts OS-level memory allocation — closing the loop between prediction and system actioning
- Designed and benchmarked multiple architectures (scikit-learn regressors, Keras sequential model, PyTorch) trained on live psutil process data (memory, CPU, priority); selected final model on MAE and latency
- Developed FastAPI backend to serve model predictions and trigger allocation adjustments programmatically

EXPERIENCE

Python Development Intern — *CodSoft (Remote)* Dec 2024 – Jan 2025

- Developed Python applications focused on data processing and automation; improved performance by 30% through optimised data structures and algorithms

- Collaborated with cross-functional team using Git and agile workflows; produced documentation and unit tests to ensure maintainability

Outreach Core Team Member — *Google Developer Groups VIT, Mumbai* Oct 2025 – Present

- Drive outreach and community engagement for 500+ student developers; organise workshops on cloud computing and data engineering with Google Developer Experts

Indigo Squad Member — *Mood Indigo, IIT Bombay* Dec 2024 – Present

- Event coordination and execution for Asia's largest college cultural festival under tight deadlines

CERTIFICATIONS & ACHIEVEMENTS

- Azure Fundamentals (AZ-900) & Azure Data Fundamentals (DP-900) — In Progress (dual certification track)
- Introduction to Machine Learning — Completed foundational ML concepts and model development
- Data Structures and Algorithms Using Java — Advanced problem-solving and algorithmic thinking
- Accenture Nordics Consultant Job Simulation — Business analysis and consulting methodologies
- PaperTrail Hackathon (March 2026) — 24-hour team project; led backend development and co-designed system architecture for SDG 16 government digitisation pipeline
- Self-directed learner: actively progressing through Apache Kafka, Apache Spark, and modern data engineering practices